PREREGISTRATION

External validation of NSGA-II electrode selection for EEG-based detection of mild cognitive impairment: a replication of Aljalal et al. (2024)

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1. Study information

Title *(required field)*

External validation of NSGA-II electrode selection for EEG-based detection of mild cognitive impairment: a replication of Aljalal et al. (2024)

Authors *(required field)*

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Description *(optional field)*

Aljalal et al. (2024, Scientific Reports 14:12483, doi:10.1038/s41598-024-63180-y) reported up to 95% accuracy in detecting mild cognitive impairment (MCI) from resting-state EEG with as few as 7 electrodes, using NSGA-II to select channels and features in a public dataset of 24 people (Isfahan, EEG_1). The authors did not release code. We rebuilt the method in Python from the published description. In the development data (EEG_1) we found three problems: (a) segment-level validation mixes segments of the same person; (b) NSGA-II selects and reports the same leave-one-subject-out accuracy, so the reported figure is optimistically biased (with shuffled labels the procedure reaches 72–79%); and (c) the frontal electrodes favoured by the paper carry substantial ocular signal.

This study tests, once and on people never used for development, whether the published approach and four alternatives (frozen before this registration) generalise. All methods are trained and select electrodes only on EEG_1 (24 people) and are evaluated once on an independent public dataset from the same centre and equipment (EEG_2, 34 people: 18 MCI, 16 controls). A secondary analysis asks whether the electrode savings hold when each electrode is processed on its own (physical montage), as a device with only those electrodes would require.

Hypotheses *(required field)*

Primary measure: **person-level balanced accuracy** (mean of sensitivity and specificity) on the 34 people of EEG_2. Non-inferiority margin: **−10 percentage points**, the smallest loss we consider relevant for screening; it is also about the size of the gain that artefact cleaning gave in development (≈9 points).

	Prediction	**Confirmed if**
P (no directional prediction)	Open question: which methods perform above chance?	Label-permutation $p < 0.05$ after Holm correction across the 5 methods (R, M0, M4, M1, M6)
H1	The paper's replication (R) will show high internal accuracy but much lower accuracy on EEG_2	R's internal accuracy (the figure the paper would report) lies above the upper limit of the 95% CI of its external balanced accuracy
H2	Clean EEG without selection (M0) will match or exceed R	Lower limit of the 90% bootstrap CI of (M0 − R) > −10 (one-sided test, $\alpha = 0.05$)
H3	Parsimonious methods (M4, M6) will match or exceed the paper's selection rule (M1) with fewer electrodes	For each: non-inferior to M1 (margin −10, Holm across the 2 comparisons) AND median number of electrodes lower than M1's
H4 (secondary)	M0 with physical montage will not be inferior to M0 with ICA. No prediction for M4 with physical montage (it fell to 36.7% in development; see section 6)	For each of M0 and M4: lower limit of the paired bootstrap CI of (physical − ICA) > −10, Holm across the 2 comparisons

Why H4. M0 and M4 compute each electrode after an average reference and an ICA fitted on all 19 electrodes. A device with only 4 electrodes could not compute that signal, so a claim of "4 electrodes" would refer to virtual electrodes. H4 tests whether the conclusion holds with real electrodes. It is secondary because it was added after the main methods were frozen and has lower power (section 3).

2. Design plan

Study type *(required field)*

Other: secondary analysis of existing public EEG data; external validation of diagnostic classification methods (development cohort → independent test cohort).

Blinding *(required field)*

No blinding. Additional blinding: no model has been trained or evaluated on EEG_2 and no metric has been computed with its labels. The test is run by frozen scripts that read the labels only at evaluation time and apply the decision rules automatically (`analisis_prereg.py`).

Study design *(required field)*

Two-cohort design with a single, locked external test. Development cohort: EEG_1 (Isfahan, 24 people: the original paper also excluded 3 controls without naming them; we exclude the 3 with most segments above 100 µV). Test cohort: EEG_2 (Isfahan, 34 people, no exclusions). Every method is fitted and selects its electrodes on EEG_1 only, then predicts each EEG_2 person once. Within-

method comparisons are paired (same 34 people). Order of execution, run once after this registration is submitted:

```
python -m mci_nsga2.externo probar --train externo/EEG_1.npz --test
externo/EEG_2.npz
python -m mci_nsga2.fisico probar --train externo/EEG_1_fisico.npz --test
externo/EEG_2_fisico.npz \
       --ica-train externo/EEG_1.npz --ica-test externo/EEG_2.npz
python -m mci_nsga2.analisis_prereg
```

Randomization *(optional field)*

No randomisation of participants. All stochastic steps use fixed seeds: NSGA-II for R seed 0; M1 and M6 seeds 0–4; ICA random_state 0; permutation and bootstrap generators seed 0.

3. Sampling plan

Existing data *(required field)*

Registration prior to analysis of the data.

Explanation of existing data *(optional field)*

EEG_1 (development) has been fully analysed; results already known are listed in section 6. EEG_2 (test) was downloaded and used only to: (a) count people per group (18 MCI, 16 controls); (b) check recording properties (all 34 have the 19 channels at 256 Hz and last 27.3–37.9 min); (c) check that no file is identical to an EEG_1 file (MD5) and that no 30-s fragment duplicates an EEG_1 recording (maximum correlation 0.74); and (d) extract features with the frozen code (`externo extraer` and `fisico extraer`), which stores the labels in the feature file without fitting any model.

The extracted feature files are fixed by their SHA-256 fingerprints. Extraction is deterministic: running it again on the same recordings reproduces these files bit for bit (verified for EEG_1 in two independent runs), so anyone can check that the registered data were used.

File	Content	SHA-256
EEG_2.npz	Test, ICA montage and replication features	cc9236c74d3797e5c10ad7d0b511951495460435e993f2d801ccb9f3b1a5a7d5
EEG_2_fisico.npz	Test, physical montage	1d829b7a383ac4e74feacf409b2cc66ea850a14cfc40c68892c2c882bff18066
EEG_1.npz	Development, ICA montage and replication features	5a4cee0f3558edd97ff696ef9ce82700a47eb593703bf04488d0e8550d16f228
EEG_1_fisico.npz	Development, physical montage	f19a619264d0785fbcedf093efe34ffadc7f7dd3f4693f063b4db2df99a31e0c

Data collection procedures *(required field)*

No new data are collected. Both datasets are public, from Isfahan University of Medical Sciences, recorded with the same equipment (19 channels, 10–20 system, 256 Hz, resting state, about 30 min) and distributed at misp.mui.ac.ir/en/eeg-data.

	EEG_1 (development)	EEG_2 (test)
People	27 (11 MCI, 16 controls)	34 (18 MCI, 16 controls)
Used	24: the original paper excluded 3 controls without naming them; we exclude the 3 with most segments > 100 µV (HC_12, HC_17, HC_24)	34: no exclusion for signal quality
Diagnosis (as described by the source)	Neuropsychiatric interview based on Petersen criteria; MMSE 21–26 for MCI, > 26 for controls; ages 60–77; coronary angiography in the previous year	Released as an extension of the same dataset; we could not verify that inclusion criteria are identical
Reference	Kashefpoor, Rabbani & Barekatain (2016), J Med Signals Sens 6:25–32	Kashefpoor, Rabbani & Barekatain (2019), Biomed Signal Process Control 53:101559

Labels come from each dataset's `states.xlsx` file, read automatically without manual review.

Sample size *(required field)*

Development: 24 people (11 MCI, 13 controls). Test: 34 people (18 MCI, 16 controls).

Sample size rationale *(optional field)*

The size is fixed by the available data; no other external data from the same equipment exist. With 34 people (18/16), a method needs at least 62% balanced accuracy to beat chance (95th percentile of the null, 20,000 simulations). Power to beat chance is 0.40, 0.63, 0.84, 0.95 and 0.99 for true balanced accuracies of 60, 65, 70, 75 and 80%. For H4, if both montages have 60–70% true accuracy and disagree on 20–35% of people, power to show non-inferiority with margin −10 is 0.44–0.70. A 24/34 result (70.6%) has a 95% CI of 52.5–84.9%. Non-confirmation will therefore be reported as inconclusive, not as evidence of no effect.

Stopping rule *(optional field)*

Not applicable: the test is run once on the complete EEG_2 dataset.

4. Variables

Manipulated variables *(optional field)*

None at the participant level. The compared conditions are the analysis methods (R, M0, M4, M1, M6) and, for H4, the montage (ICA vs physical) of M0 and M4.

Measured variables *(required field)*

- Diagnosis: MCI (1) or control (0), from `states.xlsx`.
- Resting-state EEG, 19 channels, first 10 minutes, 60 segments of 10 s per person.
- Each method's predicted diagnosis for each EEG_2 person; for M0 and M4 also the continuous LDA score.
- Number of electrodes each method uses.

Indices *(optional field)*

Balanced accuracy = (sensitivity + specificity) / 2, computed on the 34 test people. Also reported: people correctly classified out of 34, accuracy with exact (Clopper–Pearson) 95% CI, sensitivity, specificity, and AUC where a continuous score exists. For M1 and M6 each person's prediction is the **majority vote of the 5 seeds**; individual seeds are also reported.

5. Analysis plan

Statistical models *(required field)*

The five main methods were frozen in `persona.py` and `externo.py` before this registration. The physical montage (`fisico.py`) repeats M0 and M4 changing only the montage.

Method	Signal	Features	Electrode selection (on EEG_1 only)	Classifier / decision
R (paper replication)	19 channels, band-pass 0.5–64 Hz, average reference, notch 50/100 Hz	DWT db4, 5 levels without D1; threshold entropy (ThEn, 2 µV); per 10-s segment	NSGA-II, population 200, 50 generations, seed 0; objectives: fewer channels and higher segment-level LOSO accuracy; highest-accuracy solution chosen	Paper's discriminant analysis (linear or quadratic, chosen by a gene); a person is MCI if more than half of their segments are
M0	As R + ICA with 15 components (the 2 components most correlated with VEOG and HEOG removed); Fp1, Fp2, F7, F8 dropped; average reference recomputed	Log relative power (Welch, 512 points) in delta 1–4, theta 4–8, alpha 8–13, beta 13–30 Hz, averaged per person	None: 15 electrodes	Standardisation + LDA with Ledoit–Wolf shrinkage
M4	As M0	As M0	One electrode per zone (frontal, central, temporal, parieto-occipital): exhaustive search over 180 combinations, criterion inner leave-one-out balanced accuracy	As M0
M1	As M0	As M0	NSGA-II, population 40, 25 generations, paper's rule (maximum inner accuracy); seeds 0–4	As M0; majority vote of 5 seeds

Method	Signal	Features	Electrode selection (on EEG_1 only)	Classifier / decision
M6	As M0	As M0	Binary PSO, 20 particles, 25 iterations; fitness = inner balanced accuracy − 1 point per electrode; seeds 0–4	As M0; majority vote of 5 seeds
M0-physical, M4-physical (H4)	Same 15 electrodes; band-pass 0.5–64 Hz and notch 50/100 Hz per channel; no average reference and no ICA (original recording reference kept)	As M0	As M0 and M4	As M0

Transformations *(optional field)*

Only those in the table above. Features are standardised with statistics from EEG_1 only. No harmonisation between datasets is applied.

Inference criteria *(optional field)*

- **Against chance (P):** test-set predictions are fixed and test labels are permuted 10,000 times; p = (1 + #null ≥ observed) / 10,001. Holm correction across the 5 methods; $\alpha = 0.05$.
- **Non-inferiority (H2, H3, H4):** paired bootstrap of people, stratified by class, 10,000 replicates. Non-inferiority is concluded if fewer than 5% of replicates fall at or below −10 (equivalently, the lower limit of the 90% two-sided CI is above −10). Holm correction across the 2 comparisons of H3 and, separately, across the 2 comparisons of H4.
- **H1:** R's internal accuracy compared with the 95% bootstrap CI (stratified, 10,000 replicates) of its external balanced accuracy.
- **Interpretation:** a CI that includes both −10 and 0 is inconclusive; methods will not be called “equivalent”. Failing to show non-inferiority does not show inferiority. All methods and all seeds are reported regardless of outcome.
- **Dataset effect (descriptive):** an LDA with leave-one-out tries to tell which dataset each person comes from using only their EEG features (58 people). If its balanced accuracy is ≥ 75%, the domain shift is large and any drop on EEG_2 will be discussed as possibly due to the dataset and not only to the method.

Data exclusion *(optional field)*

EEG_1: the 3 controls with most segments above 100 µV (HC_12, HC_17, HC_24). The original paper excluded 3 controls without naming them; this rule is our reconstruction, and the analysis with all 27 people is reported as exploratory. EEG_2: no exclusion; a recording would be excluded only if the code could not read it or one of the 19 channels were missing, which has already been checked and does not occur. No outlier removal at the person level.

Missing data *(optional field)*

None expected: all 34 EEG_2 recordings exceed 10 minutes and contain all channels. If a recording yielded fewer than 60 valid segments, the available segments would be used and this would be reported.

Exploratory analysis *(optional field)*

Reported as exploratory, without confirmatory value:

- Sensitivity analysis with all 27 people of EEG_1 (no control excluded).
- Leave-pair-out AUC in EEG_1 and external AUC for every method with a continuous score.
- Mains noise: whether 50-Hz power differs between groups or datasets and whether it predicts the label.
- Residual ocular contamination after ICA and its relation to predictions.
- Positive control: whether posterior alpha power, a feature with a known effect in MCI (Babiloni et al., 2015), shows the expected direction in both datasets.
- Results of each seed of M1 and M6 and stability of the selected electrodes.

6. Other

Other *(optional field)*

6.1 Development results already known (EEG_1)

With 24 people, no method beat chance under honest evaluation. These results motivated the hypotheses and are declared so that readers know what was known before the test.

- 19 electrodes, person-level LOSO: about 53% (chance 54%); 10-fold with segments of the same person mixed: about 91%.
- NSGA-II with shuffled labels: 72–79% internal accuracy, equal to or above that with real labels (65–75%).
- Nested LOSO (DWT-ThEn-KNN): 67% internal versus 40% on the held-out person.
- ICA cleaning: about 9 points more balanced accuracy (from about 48% to 57%), $p \approx 0.14$.
- Paper's selection rule in nested LOSO: 48%, versus 61% without selection. With shuffled labels one method reached 71% even under nested evaluation.

Physical montage versus ICA, nested LOSO, 24 people (computed before this registration):

Method	Correct	Balanced accuracy	AUC	Permutation p	Most frequent selection
M0 with ICA	15/24	61.2%	0.61	0.24	15 electrodes
M0 physical	15/24	61.2%	0.57	0.23	15 electrodes
M4 with ICA	16/24	64.3%	0.61	0.11	F4, Cz, T4, P3 (5 of 24 folds)
M4 physical	9/24	36.7%	0.29	0.96	F4, T3, Cz, O2 (5 of 24 folds)

M0 gives the same accuracy with both montages, although predictions agree for only 75% of people. M4-physical falls below chance and picks different combinations in almost every fold. With n = 24

this is compatible with selection noise (below-chance leave-one-out accuracy is a known small-sample bias). Hence H4 makes no prediction for M4-physical.

6.2 Changes to the frozen code

The only change to the main methods' code is in `externo.py`: the report now stores each person's prediction and label and the channel list chosen by R. No computation changes; the paired comparisons of H2 and H3 need it. The majority vote of M1 and M6 is fixed here because the original code reports each seed separately. New files: `fisico.py` (H4) and `analisis_prereg.py` (decision rules). Details in `CAMBIOS.md`.

6.3 Deviations

Any change to code, data or rules after submission will be reported in the article with its reason and, where possible, with results of both versions. If the code fails on EEG_2 for a technical reason (file reading, channel names), it will be fixed without touching the methods and documented.

6.4 Code and fingerprints

The code package (`mci_nsga2_v1.0_codigo.zip`) is attached to this registration and will be published on GitHub with a Zenodo DOI. SHA-256 fingerprints of the files at registration (full list of 22 files in `SHA256SUMS.txt`):

File	Content	SHA-256
mci_nsga2/externo.py	External test R, M0, M4, M1, M6	a887b3227ecbfe67342068070a7cde72bf121ea7592c249477140bffa73cbfa6
mci_nsga2/persona.py	Person-level selection (zones, NSGA-II, PSO)	f8dc2c98430a80bc2a01fd91f977fe21f58f4729d6d7ffd51d4ca23ed8e4eddf
mci_nsga2/limpieza.py	ICA and relative power	5e19be2bc9bdf34f40fb228c3c991b0971609d2ce853836521c06b54587dadf7
mci_nsga2/fisico.py	Physical montage and H4 comparison	c675b90a09da39297f9f32a9530e01262f1d39ec12a0fb52b3a336d78bce9c53
mci_nsga2/analisis_prereg.py	Decision rules H1–H4	2e67f30f6680b815eea53f599578b050bdfd3cb0c9cb5b310e7f7117a2838346
mci_nsga2/nsga2.py	NSGA-II	d89d8d6529e0987eb868292810b8680d34e56961c393f0abd5822042de01abda
mci_nsga2/classify.py	Replication classifiers	f7d4b8ec5d2d1141dbb0be096e3aa2be9258982d7ff09f1d3cff24f0ddb6df5a
mci_nsga2/config.py	Constants and assumptions	d91d0de55ea31cb7376d2bec24702aa933cb63571ac72ba81241d86d8df8d8b1
mci_nsga2/data_loader.py	EDF reading and preprocessing	577e99c094759397bb9b83cce4195569976f80d38ec66abec3d254198212807f
mci_nsga2/features.py	Replication features	506f3497946fb179f12c258e11ec34216db077d54ada16384391fee540c94d3f

Package file: mci_nsga2_v1.0_codigo.zip — SHA-256
1c18b3a2f721af15f31ec38c0b0b0d1f9fc78e9c1afaa7ce112cf387017793ed.

6.5 References

Aljalal, M., Aldosari, S. A., Molinas, M., & Alturki, F. A. (2024). Selecting EEG channels and features using multi-objective optimization for accurate MCI detection: validation using leave-one-subject-out strategy. Scientific Reports, 14, 12483. https://doi.org/10.1038/s41598-024-63180-y

Babiloni, C., et al. (2015). Occipital sources of resting-state alpha rhythms are related to local gray matter density in subjects with amnesic mild cognitive impairment and Alzheimer's disease. Neurobiology of Aging, 36, 556–570. https://doi.org/10.1016/j.neurobiolaging.2014.09.011

Kashefpoor, M., Rabbani, H., & Barekatain, M. (2016). Automatic diagnosis of mild cognitive impairment using electroencephalogram spectral features. Journal of Medical Signals and Sensors, 6(1), 25–32.

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